(4) (i) $5\tan\frac{x}{5}+c$ (ii) $-\frac{1}{5}\operatorname{cosec}(5x+3)+c$ (iii) $-\frac{1}{15}\sec(2-15x)+c$

(5) (i) $\frac{1}{4}\sin^{-1}(4x)+c$ (ii) $\frac{1}{9}\sin^{-1}(9x)+c$ (iii) $\frac{1}{6}\tan^{-1}(6x)+c$

Exercise 11.3

(1) $\frac{(x+4)^6}{6}+\frac{1}{3(2-5x)^3}+\frac{\cot(3x-1)}{3}+c$ (2) $-2\sin(5-2x)+3e^{3x-6}-6\log|6-4x|+c$

(3) $5\tan\frac{x}{5}+9\sin 2x+2\sec(5x+3)+c$ (4) $2\sin^{-1}(4x)+9\sin^{-1}(3x)-3\tan^{-1}(5x)+c$

(5) $2\tan^{-1}(3x+2)+3\sin^{-1}(3-4x)+c$ (6) $\sin\left(\frac{x}{3}-4\right)+\log|7x+9|+5e^{\frac{x}{5}+3}+c$

Exercise 11.4

(1) $2x^2-5x+3$ (2) $3(x^3-x^2-1)$ (3) $2x^3-3x^2+5x+26$

(4) (i) 8 seconds (ii) 39.2 m/sec (iii) 78.4 m/sec

(5) (i) 2.4 sq.cm (ii) 0.4 sq.cm

Exercise 11.5

(1) $\frac{x^2}{2}+4x-3\log|x|-\frac{2}{x}+c$ (2) $\frac{x^2}{2}+\log|x|+2x+c$ (3) $\frac{8x^3}{3}+26x^2-180x+c$

(4) $\tan x-\cot x-2x+c$ (5) $2[\sin x+x\cos\alpha]+c$ (6) $-2\operatorname{cosec}2x+c$

(7) $-3\cot x-4\operatorname{cosec}x+c$ (8) $x-\sin x+c$ (9) $2\left[\frac{\sin 3x}{3}+\sin x\right]+c$

(10) $\frac{1}{2}\left[\frac{\sin 5x}{5}+\sin x\right]+c$ (11) $\frac{1}{2}\left[x-\frac{\sin 10x}{10}\right]+c$ (12) $-\frac{1}{8}\cos 4x+c$

(13) $\frac{(ae)^x}{\log(ae)}+c$ (14) $\frac{2}{15}(3x+7)^{\frac{5}{2}}-\frac{2}{3}(3x+7)^{\frac{3}{2}}+c$

(15) $\frac{2^{2x+2}}{\log 2}-\frac{2^{2-3x}}{3\log 2}+c$ (16) $\frac{2}{21}[(x+3)^{\frac{3}{2}}+(x-4)^{\frac{3}{2}}]+c$

(17) $2\log|x+3|-\log|x+2|+c$ (18) $\frac{1}{9}\log|x-1|-\frac{1}{9}\log|x+2|+\frac{1}{3(x+2)}+c$

(19) $\log\left|\frac{x+2}{x-1}\right|+3\tan^{-1}x+c$ (20) $\frac{x^2}{2}+3x-\log|x-1|+8\log|x-2|+c$

Exercise 11.6

(1) $\sqrt{1+x^2}+c$ (2) $\frac{1}{3}\tan^{-1}(x^3)+c$ (3) $\log|e^x+e^{-x}|+c$

(4) $\log|10^x+x^{10}|+c$ (5) $-2\cos\sqrt{x}+c$ (6) $\log|\log(\sin x)|+c$

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